Re-Align

A Novel Prosthesis for Hip Dysplasia

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PATIENT PROFILE

Djimon Hounsou

41 years old | Male

Weight: 85 kg | Height: 185cm

Symptoms

- Bilateral hip pain
- Reduced ability to walk for prolonged periods of time
- Frequent hip dislocations since birth
- Limited range of motion

Diagnosis: Developmental Hip Dysplasia

- Connection between femoral head and acetabulum is shallow in right hip

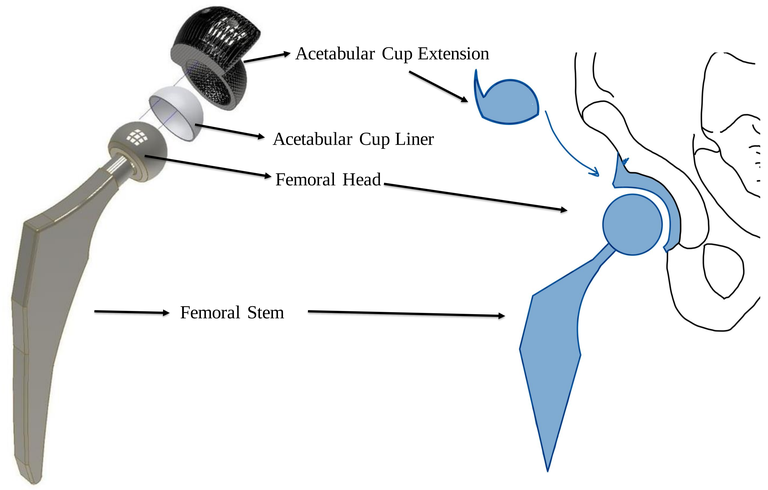


FORM

- Flat tapering, wedge geometry of implant enhances proximal joint offloading, bone preservation and provides rotational stability [1]
- Larger stem size for maximized bone growth and healing around stabilized surface
- Pelvic extension added to acetabular cup to fully restructure and align joint
- Porous hydroxyapatite coating to encourage bone ingrowth and healing

Dimensions	Values
Joint Cup Diameter	44mm
Stem Length	220mm
Offset Angle	124°

OUR DESIGN



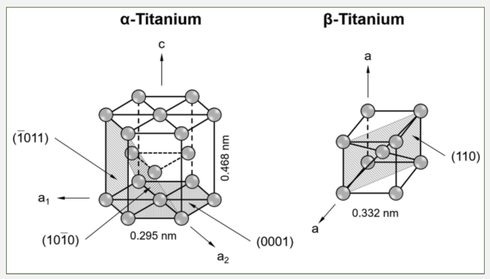
Justification of design

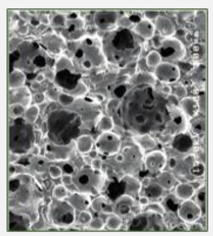
- Total Hip Replacement to repair compromised hip joint
- Metal-on-Polyethylene implant for joint stability, low friction, and smooth movement
- Extended acetabular cup to provide strength and area to thin acetabulum

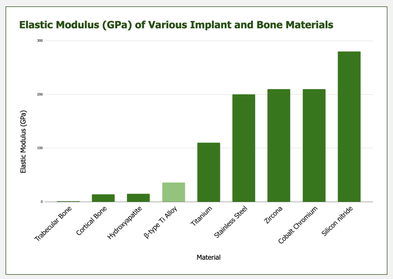
MATERIALS

β-type Titanium Alloy Ti-6Al-7Nb

- High ultimate strength for acetabular component and femoral shaft (1063 MPa) [2]
- β-phase: softer lattice structure to reduce stress shielding effect [3]
- Porous hydroxyapatite coating for bone ingrowth
- Biocompatibility, low cytotoxicity, corrosion and wear resistance

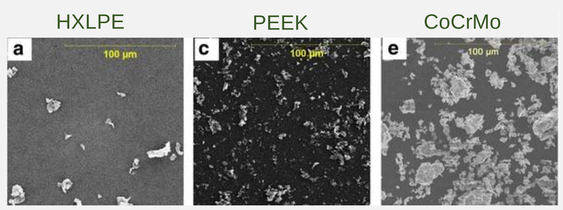


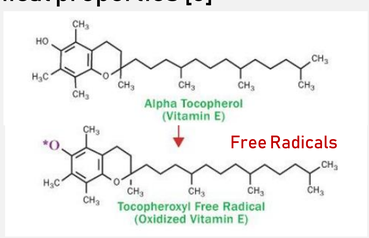




Vitamin E-stabilized Highly Cross Linked (HXLPE) Polyethylene Liner

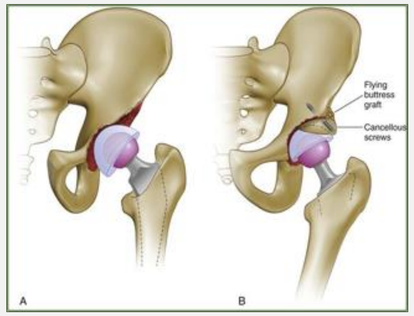
- Cross-linked polymer reduces friction and protects from metallosis [4], [5]
- Developed to reduce wear in-vitro and debris-induced osteolysis [4]
- Vitamin E removes free radicals without sacrificing mechanical properties [6]





SURGICAL FIXATION

- Femur is shortened to even out legs
- Secure acetabular cup using screws and bone ingrowth will fixate it [7]
- Autograft from femur is used to fill gaps in acetabulum and promote bone growth [8]
- Femoral stem is inserted into top of thigh bone and fixate by osteointegration [9]



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